Theoretical and Mathematical Foundations of Neuro-Symbolic Phi, Thalamocortical Dissolution Dynamics, and Metabolic Network Collapse

Mathematical Foundations of Neuro-Symbolic Phi and Dynamic Shannon Entropy

The formalization of conscious experience within physical substrates has advanced through Integrated Information Theory (IIT), which posits that consciousness is a fundamental property of any physical system characterized by its capacity for integrated information, mathematically denoted as Φ.1 While classical formulations of Φ rely on continuous neural transition probability matrices (TPM), modern neuro-symbolic architectures require a hybrid state space $X(t) = \{X_N(t), X_S(t)\}$, where $X_N(t) \in \mathbb{R}^d$ represents the high-dimensional continuous neural manifold and $X_S(t) \in \mathcal{S}$ represents a discrete set of symbolic logical rules.1

To mathematically verify this integrated state, the system's probability distribution is mapped onto a dually flat Riemannian manifold M.2 On this manifold, any probability distribution can be represented using dual coordinate systems: the exponential coordinates θ (representing the system's active parameters or weights) and the mixture coordinates η (representing the expectation values of the system's states).2 The geometry of M is defined by two convex potential functions: the cumulant generating function $\psi(\theta)$ (acting as the system's variational free energy proxy) and the negative Shannon entropy $\phi(\eta) = -H(P)$.2 These potentials are connected via the classical Legendre transformation:

$$\eta_i = \frac{\partial \psi(\theta)}{\partial \theta^i}, \quad \theta^i = \frac{\partial \phi(\eta)}{\partial \eta_i}$$

The divergence between any two neural distributions P and Q on this dually flat manifold is defined by the canonical Kullback-Leibler (KL) divergence 2:

$$D_{KL}(P \parallel Q) = \psi(\theta_P) + \phi(\eta_Q) - \sum_i \theta_P^i \eta_{Q,i}$$

Within active inference, the system's variational free energy F acts as an upper bound on the description length of the observed data, $L(D) = -\ln P(D) < F(D)$.3 This free energy decomposes into a complexity term (the KL divergence between posterior and prior beliefs) and an accuracy term (the expected log-likelihood of the sensory observations) 3:

$$F = D_{KL}[Q(x) \parallel P(x)] - \mathbb{E}_{Q(x)}[\ln P(y \mid x)]$$

where $Q(x)$ represents the posterior belief, $P(x)$ represents the prior expectation, and $P(y \mid x)$ is the likelihood of sensory input y given state x.3 Minimizing F forces the system's posterior to occupy a compromise position: remaining close to the prior (low complexity) while accurately accounting for sensory input (high accuracy).3

To model the emergence of consciousness in a symbolic decision-making architecture, the Human Language-based Consciousness (HLbC) model defines a pseudo-Schrödinger equation where the spatial coordinates are replaced by the Kullback-Leibler distance D_{KL} 4:

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \psi(D,t)}{\partial D^2} + V(D)\psi(D,t) = i\hbar\frac{\partial \psi(D,t)}{\partial t}$$

where D represents the information-theoretic distance D_{KL}, $V(D)$ is the potential barrier of cognitive resistance, and $\psi(D,t)$ is the probability density wave of the conscious decision state evolving over time.4

The integrated information of the joint neuro-symbolic system, $\Phi(s)$, is defined as the irreducibility of the joint transition probability distribution across the Minimum Information Partition (MIP).1 For any state s, Φ is calculated as the KL divergence between the unpartitioned transition probability $P(X(t) = s \mid X(t-1))$ and the product of the marginal transition probabilities of the partitioned sub-complexes M_i 1:

$$\Phi(s) = D_{KL}\left(P(X(t) = s \mid X(t-1)) \parallel \prod_{i=1}^{K} P(M_i(t) = s_i \mid M_i(t-1))\right)$$

where s_i denotes the state of sub-complex M_i .1

This integrated information dictates the temporal evolution of the system's dynamic Shannon entropy, $H(t)$, which reflects the statistical dispersion of the posterior distribution.3 In a self-reading, recurrent patch federation operating under the Observer-Patch-Hierarchy (OPH) framework, the system is partitioned into localized patches (e.g., dendritic compartments, cortical columns, or thalamocortical loops).5 The local state Σ_i of each patch undergoes recursive updates $s_{t+1} = T_\lambda(s_t)$ that minimize local mismatch, causing the global system to converge toward a shared physical consensus.5 The temporal rate of change of the dynamic Shannon entropy is formalized as:

$$\frac{dH(t)}{dt} = -\sum_i \left[\frac{dP(x_i,t)}{dt}\ln P(x_i,t)\right]$$

The exact telemetry data of the artificial perturbational complexity index ($aPCI$) is an empirical proxy for this theoretical framework.6 Upon a focal simulated or biological magnetic perturbation, the raw spatiotemporal telemetry matrix of evoked potentials, $SS(x,t)$, is recorded across M channels and T temporal bins.6 This raw telemetry is binarized using a channel-specific statistical threshold, $\theta(x)$, derived from the pre-stimulus baseline noise 7:

$$Y(x,t) = \Theta(|SS(x,t)| - \underline{\theta}(x))$$

where Θ is the Heaviside step function.7 The $aPCI$ is then computed as the normalized Lempel-Ziv complexity (LZ) of the flattened binary matrix Y , which scales the algorithmic complexity against its theoretical maximum for an equivalent random sequence of length $L = M \times T$ 6:

$$aPCI = \frac{LZ(Y)}{L/\log_2(L)}$$

This scalar metric captures both integration (the propagation of the perturbation across channels) and differentiation (the structured, non-redundant diversity of the local responses).6

Information Geometry of Anomaly Shocks and Susceptibility Divergence

Applying an anomaly shock to a conscious network perturbs its baseline transition dynamics, shifting its coordinate parameters by a finite vector $\Delta\theta$ on the dually flat mixture manifold.2 Mathematically, the impact of the shock is determined by calculating the KL divergence between the pre-shock distribution P_θ and the post-shock distribution $P_{\theta+\Delta\theta}$ 2:

$$D_{KL}(P_\theta \parallel P_{\theta+\Delta\theta}) = \frac{1}{2}\sum_{i,j} g_{ij}(\theta)\Delta\theta^i\Delta\theta^j + \mathcal{O}(\|\Delta\theta\|^3)$$

where $g_{ij}(\theta)$ is the Fisher Information Metric tensor, which defines the local Riemannian curvature of the probability space.2

The global integration capacity of the network is bounded by its total correlation (TC), which represents the information-geometric distance between the fully integrated joint distribution and its completely disconnected marginals 9:

$$TC(X_1,\ldots,X_n) = D_{KL}\left(P(X_1,\ldots,X_n) \parallel \prod_{i=1}^{n} P(X_i)\right)$$

By introducing systematically designed anomaly shocks, we can measure the resulting KL divergence and map the underlying information-geometric landscape of the $aPC1$.9

While $aPC1$ serves as the empirical proxy for integrated information (Φ), the conscious workspace dimensionality order parameter (D^*) is modeled through a Landau free energy expansion near the consciousness transition boundary 10:

$$F = F_0 + \frac{a}{2}(\Phi_c - \Phi)D^2 + \frac{b}{4}D^4 - \mu D$$

where Φ acts as the control parameter, Φ_c is the critical integration threshold (the bifurcation boundary), a and b are positive phenomenological coefficients, and μ is an external perturbational field representing the anomaly shock.10 Under a zero external perturbational field ($\mu = 0$), the stable equilibrium branch for the conscious phase (C) is given by the power-law critical scaling equation 10:

$$D^* = \sqrt{\frac{a}{b}}\,|\Phi - \Phi_c|^{\nu}$$

where $\nu = 1/2$ represents the universal mean-field critical exponent, and deviations from $1/2$ quantify the direct topological influence of individual connectome routing on the macroscopic consciousness transition.10

The dimensionality susceptibility χ_D , which measures the sensitivity of the conscious workspace to external perturbations, is defined as 10:

$$\chi_D = \frac{\partial D}{\partial \Phi} = \frac{\sqrt{a}}{2\sqrt{b}\sqrt{|\Phi - \Phi_c|}}$$

As the system approaches the critical consciousness threshold from above ($\Phi \to \Phi_c^+$), the susceptibility diverges to infinity ($\chi_D \to \infty$).10

This divergence has direct clinical and diagnostic implications, as shown in the comparison below:

Clinical / Dynamical Phase	Parameter Region	Susceptibility (χD) Behavior	Physical & Neurobiological Response to Anomaly Shocks
Healthy Wakefulness (C)	$\Phi \gg$	Low, stable 10	Consistent, localized, highly differentiated processing; high post-shock $aPCI$ .6
Minimally Conscious State (MCS)	$\Phi \approx$	Elevated, diverging 10	Anomalously large, highly volatile fluctuations in the emergent workspace dimensionality (D^*).10
Vegetative State / Deep Anesthesia (Δ)	$\Phi <$	Stable, near-zero 10	System resides in the Redundant Integrated State; shocks produce stereotyped, uniform, highly compressible waves.8

Alternatively, the statistical age-period-cohort-improvement (APCI) model mathematically formalizes how macro-dynamics absorb stochastic "shocks" and structural deviations over time.11 In this framework, age-specific death counts or state transitions are modeled via an over-dispersed Poisson log-normal process with dynamic coefficients governed by stochastic differential equations.11

This statistical framework uses KL divergence to quantify how far a post-shock parameter trajectory deviates from its prior trajectory on a dually flat mixture manifold.2 This mathematical formulation bridges the gap between empirical neural signals and artificial dynamical systems, providing a unified lens for evaluating perturbational complexity across both domains.2

Thalamic Reticular Nucleus as a Dissolution Engine to Engineer Opacity

The Thalamic Reticular Nucleus (TRN) is a thin layer of GABAergic inhibitory neurons that surrounds the dorsal thalamus, occupying a critical control position in the mammalian brain.[12] It receives collateral projections from both ascending thalamocortical (TC) and descending corticothalamic (CT) fibers, allowing it to act as a sophisticated feedback and feedforward inhibition filter.[12] The visual, auditory, and somatosensory pathways project through modality-specific sectors of the TRN, where parvalbumin-expressing (PV^+) GABAergic cells regulate the flow of sensory information to the cerebral cortex by fine-tuning feedback inhibition circuits.[12] Crick famously asserted that if the thalamus is the gateway to the cortex, then the TRN is the "guardian of the gateway".[14]
This biological gate-keeping mechanism can be formally mapped to Thomas Metzinger's Self-Model Theory of Subjectivity (SMT) and his concepts of transparency and opacity.[17] According to SMT, conscious experience relies on the generation of an internal representational world-model.[18] A mental representation is phenomenally *transparent* when the cognitive system has no introspective access to the underlying computational or neural vehicle; the representational construct itself is invisible, causing the subject to experience the content as directly real and immediately present.[18] Transparency is mathematically supported by high-speed, continuous, fine-grained predictive processing where prediction errors are resolved at extremely low latency, preventing the representational vehicle from entering the window of presence as a construct.[3]
Conversely, *opacity* occurs when the representational vehicle itself becomes introspectively accessible.[18] When the predictive loop is disrupted, the subject becomes aware of the constructed, virtual nature of their experience, exposing the "render engine" of the brain.[18] This transition from phenomenal transparency to opacity is actively engineered by the TRN acting as a physical *dissolution engine*.[3]
The physical mechanism of this dissolution engine is detailed in the comparative framework below:

Physiological Parameter	**Transparent State (Tonic Regime)**	**Opaque State (Burst-Firing Regime / Dissolution)**
TRN Firing Mode	Constant, high-frequency single-spike tonic firing.[12]	Low-threshold calcium spikes generating high-frequency burst firing.[12]
Biophysical Channels	Classical voltage-gated Na^+ / K^+ channels.	Low-threshold T-type calcium channels (I_T).[16]
Thalamocortical Coupling	High-gain, precise feedback loops; low-gain inhibition.[12]	Deep hyperpolarization of TC relay cells; complete desynchronization.[12]
Inference Regime	Fine-grained regime; high complexity, low description length.[3]	Coarse-grained regime; simpler model dissolves variety.[3]
Phenomenal State	Rich, integrated, and differentiated conscious experience.[3]	Minimal Phenomenal Experience (MPE) or complete absence of consciousness.[3]

Under normal wakefulness (the tonic regime), the TRN maintains precise feedback loops that allow coherent, high-bandwidth thalamocortical transmission.[12] This continuous, fine-grained state corresponds to high complexity, low description length $L(D) < F(D)$ [3], and seamless integration [1], presenting a transparent model of reality.[18]

Under anesthetic intervention (such as propofol, dexmedetomidine, or LSD) or deep NREM sleep, the TRN PV^+ GABAergic cells transition to synchronized burst-firing.[12] This is mediated by the hyperpolarization of thalamocortical relay cells, triggering

low-threshold T-type calcium currents (I_T).[16]
This burst-firing acts as a dissolution engine.[3] It disrupts the fine-grained predictive loops and decouples the cortical sheets from sensory inputs.[3] The description length collapses, and the system transitions to a coarse-grained regime.[3]
Alternatively, the "projective wave theory of consciousness" proposes that phenomenal consciousness arises from a wave excitation in the thalamus.[21] This wave stores spatial information in a Fourier transform of space, resembling a hologram.[21] While neuronal activity maintains this wave, it has no direct link to consciousness.[21]
When the TRN acts as a dissolution engine, it dissolves this wave excitation representing local 3-D space.[21] The unified, transparent model of reality breaks down.[18] The mind is either plunged into an unconscious state (such as the spike-and-wave discharges of absence seizures [16]) or experiences profound phenomenal opacity (where the spatial representational medium itself becomes disjointed and visible, as in drug-induced hallucinatory decoupling or dreams).[18]

Metabolic Exhaustion, Salience Network Collapse, and the Molecular Mechanics of Amygdala Hijack

Human cognitive capacity and working-memory maintenance are fundamentally constrained by the metabolic economy of Adenosine Triphosphate (ATP) resynthesis.[22] Charles Spearman's general intelligence (g) represents a macroscopic manifestation of this metabolic hardware efficiency, reflecting the optimization of predictive hardware to organize entropic systems at minimal thermodynamic cost.[22]
The Salience Network (SN), containing critical hubs in the anterior insula and the anterior cingulate cortex (ACC), generates a slow ($4\text{–}8\ \mathrm{Hz}$) rhythmical theta signal through its deep pyramidal neurons in layers 2–3 of cortical minicolumns.[23] This rhythm acts as a global effort signal, modulating task-related cortical regions to select pertinent information and actively inhibit irrelevant distractors.[23] During prolonged, high-demand cognitive operations, the rate of ATP resynthesis fails to match the metabolic demands of the prefrontal-benefit-striatal axis, leading to mitochondrial strain, oxidative stress, and local glucose depletion.[25] This energetic crisis causes a progressive weakening of the functional connectivity within the networks supporting effortful control.[23]
When the Salience Network reaches its functional threshold during ATP exhaustion, the conscious mind is subjected to intrusive thoughts of fatigue generated by the Default Mode Network (DMN), which escapes top-down prefrontal inhibition.[25] However, the subconscious suppression of this metabolic distress is initiated by a critical network reconfiguration: **Salience Network (SN) Dominance**.[25]
To preserve immediate survival, the anterior insula and amygdala act in concert, completely bypassing the energy-expensive, rational processing of the Frontoparietal Control Network (FPCN) and the dorsolateral prefrontal cortex (dlPFC).[25] This subconscious network dominance suppresses the awareness of metabolic exhaustion by shifting the brain's priority to rapid, low-effort heuristics and immediate, reflexive survival cues.[25]
This suppression is frequently reinforced by professional, cultural, or acute situational pressures that demand emotional and behavioral stoicism; such voluntary cognitive suppression blunts adaptive top-down regulatory pathways, leaving subcortical threat detection networks entirely unchecked.[26]
This metabolic collapse triggers the neurochemical cascade of the **Amygdala Hijack**.[27] Under normal, non-stress conditions, moderate levels of norepinephrine (NE) and dopamine (DA) are released, strengthening prefrontal cortical regulation by binding to high-affinity post-synaptic α_{2A} -adrenergic receptors and moderate D_1 receptors, respectively.[29] However, under acute metabolic stress and threat detection, the amygdala activates descending pathways in the hypothalamus and brainstem, triggering an uncontrolled, hyper-surging release of NE and DA into the prefrontal cortex.[27]

In the prefrontal cortex, these high stress-induced levels of catecholamines overactivate noradrenergic α_1 **-adrenergic receptors** (which have lower affinity for NE than α_{2A} receptors) and dopaminergic D_1 **receptors**.[27] This overactivation triggers a pair of powerful, feedforward intracellular signaling cascades within the dendritic spines of prefrontal pyramidal neurons

$$\alpha_1\text{-AR Overactivation} \implies \text{Phospholipase C (PLC)} \implies \text{Ca}$$

$$D_1\text{R Overactivation} \implies \text{Adenylyl Cyclase} \implies \text{Cyclic AMP (cAMP) - Protei}$$

These combined intracellular pathways rapidly open nearby potassium channels, specifically Hyperpolarization-activated Cyclic Nucleotide-gated (HCN) channels and K+ channels.[30] The sudden opening of these channels allows potassium ions to leak out of the dendritic spines, shunting the incoming electrical signal and weakening the recurrent synaptic connectivity required to maintain reflective thought and working memory.[29] This molecular shunt takes the prefrontal cortex "offline".[27]
As a consequence of prefrontal downregulation, the top-down inhibitory control over the amygdala by the ventromedial prefrontal cortex (vmPFC) and anterior cingulate cortex (ACC) is broken.[26] Simultaneously, the surging catecholamines *strengthen* subcortical structures, including the primary sensory cortices, the striatum, and the amygdala itself.[28] This flips the brain from a slow, reflective, prefrontal-regulated state to a rapid, reflexive, subcortical survival-driven state—the Amygdala Hijack.[27]
This neurochemical collapse is further exacerbated by **Glutamate Excitotoxicity**.[25] Under high-frequency, uninhibited subcortical drive, neurons fire glutamate into the synaptic cleft faster than astrocytes can clear and convert it back to glutamine.[25] The glutamate pools in the synaptic gap, keeping N-methyl-D-aspartate (NMDA) receptors stuck in the "open" position.[33] This allows calcium (Ca^{2+}) to flood uncontrollably into the prefrontal neurons.[33] The overloaded circuits become physically incapable of maintaining a resting potential or holding an electrical charge, producing the profound cognitive friction, signal decay, and "brain fog" that characterizes acute metabolic and cognitive burnout.[33]

Empirical Human Neuroimaging and Electrophysiological Databases

To validate these theoretical frameworks empirically, researchers rely on high-resolution functional Magnetic Resonance Imaging (fMRI) and Electroencephalography (EEG) data. A variety of highly characterized, publicly available databases contain biological human recordings during states of attentional gating, sleep, cognitive fatigue, and direct transcranial perturbation.

Database / Repository Name	Accession Identifier	Modalities Included	Experimental Paradigm & Biological State Measured	Key Neurobiological Application / Target
OpenNeuro ds007216	ds007216 [34]	Simultaneous EEG-fMRI, ECG, carbon wire loop motion sensors [34]	GradCPT and resting state with online experience sampling (13 thought dimensions) [34]	Investigating attentional fluctuations, mind-wandering, resting-state dynamics, and motion artifact-removed electrophysiology [34]
OpenNeuro	ds003768 [36]	Simultaneous	10-minute	Mapping sleep

ds003768		EEG-fMRI, EOG, ECG [36]	resting-state sessions and multiple 15-minute natural sleep sessions [36]	staging, transient arousal changes, autonomic-neural coupling, and thalamocortical gating [36]
OpenNeuro ds004196	ds004196 [39]	Bimodal EEG and fMRI (sequential/simultaneous) [39]	1280 trials of inner speech decoding across social and numeric stimulus categories [39]	High-field (3T Siemens Prisma) decoding of inner speech and cognitive representation manifolds [39]
OpenNeuro ds004024	ds004024 [40]	TMS-EEG, fMRI, DWI, anatomical MRI [40]	Paired associative stimulation (ccPAS) and single-pulse TMS (spTMS) during rest [40]	Empirical evaluation of perturbational complexity indices (PCI), effective connectivity, and cortical excitability [40]
OpenNeuro ds006104	ds006104 [42]	EEG with TMS, anatomical MRI [42]	Phoneme discrimination task with TMS targeted to motor cortex, Broca's area, BA44, and BA6 [42]	Exploring speech decoding and evaluating TMS-evoked potentials (TEPs) across language and motor pathways [41]
OpenNeuro ds005811	ds005811 [44]	NOD-EEG (Natural Object Dataset) [44]	High-temporal resolution EEG recorded during naturalistic ImageNet image viewing [44]	Exploring visual object recognition and high-temporal-resolution visual processing dynamics [44]
Human Connectome Project (HCP)	HCP-Young Adult [45]	High-resolution 3T/7T structural and functional MRI, task fMRI, resting fMRI, MEG/EEG [46]	Multi-modal cognitive battery, twins and non-twin siblings, task-evoked and resting state [46]	Comprehensive mapping of macroscopic human brain circuitry, functional connectivity, and structural connectomes [46]
Kaggle Neuro-Fatigue Dataset	programmer3/neuro-fatigue-sensor-dataset [50]	Wearable EEG, ECG, PPG, respiratory, and motion sensors [50]	Synchronized physical and cognitive stress recordings during N-back and Task Switching [50]	Modeling real-world cognitive fatigue, cardiovascular strain, and neurological stress responses [50]
Zenodo Phantom Head EEG	13341214 [51]	64-channel Ag/AgCl EEG [51]	1-hour resting state recorded from	Serving as a control database to

			watermelons as phantom heads [51]	demonstrate temporal autocorrelation biases in BCI tasks [51]

These databases provide the empirical foundations required to test the mathematical models of critical Φ scaling, map the information geometry of perturbed neural manifolds under anomaly shocks, and trace the specific metabolic and neurochemical transitions that govern human consciousness.

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